

DEPARTMENT OF MATHEMATICS
MAT122 INTRODUCTORY MATHEMATICS II

WEEK 9: LECTURE NOTE

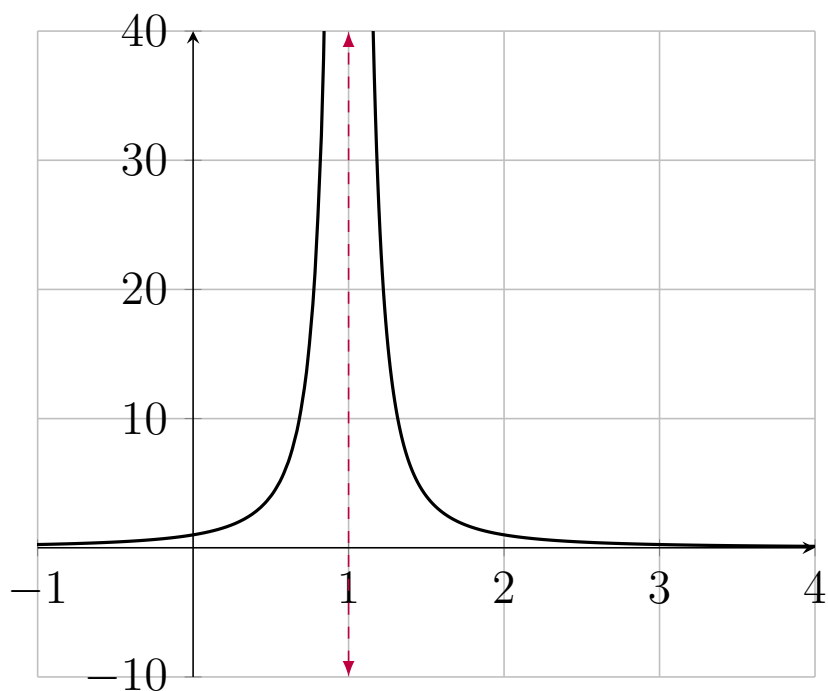
APPLICATIONS OF DIFFERENTIATION continued.....

INFINITE LIMITS

Definition 1 (Definition of Infinite Limits). *Let f be a function that is defined at every real number x in some open interval containing c (except possibly at c itself). A limit in which $f(x)$ increases and decreases without bound as x approaches c is called an **infinite limit**. The statement*

1. $\lim_{x \rightarrow c^-} f(x) = -\infty$ means that $f(x)$ decreases without bound as x approaches c from the left.
2. $\lim_{x \rightarrow c^-} f(x) = +\infty$ means that $f(x)$ increases without bound as x approaches c from the left.
3. $\lim_{x \rightarrow c^+} f(x) = -\infty$ means that $f(x)$ decreases without bound as x approaches c from the right.
4. $\lim_{x \rightarrow c^+} f(x) = +\infty$ means that $f(x)$ increases without bound as x approaches c from the right.
5. $\lim_{x \rightarrow c} f(x) = -\infty$ means that $f(x)$ **decreases without bound as x approaches c** .
6. $\lim_{x \rightarrow c} f(x) = +\infty$ means that $f(x)$ **increases without bound as x approaches c** .

Example 2. Determine the limit of the function in the Figure below as x approaches 1 from the left and from the right.



It is simple observation that gives us:

$$(i) \lim_{x \rightarrow 1^-} \frac{1}{(x-1)^2} = +\infty \quad (ii) \lim_{x \rightarrow 1^+} \frac{1}{(x-1)^2} = +\infty \quad \lim_{x \rightarrow 1} \frac{1}{(x-1)^2} = +\infty$$

Vertical Asymptotes

Definition 3. If $f(x)$ approaches infinity (or negative infinity) as x approaches c from the right or the left, then the vertical line $x = c$ is called a **vertical asymptote** of the graph of f .

Theorem 4 (Vertical Asymptotes). Let f and g be continuous on an open interval containing c . If $f(c) \neq 0$, $g(c) = 0$, and there exists an open interval containing c such that $g(x) \neq 0$ for all $x \neq c$ in the interval, then the graph of the function

$$h(x) = \frac{f(x)}{g(x)}$$

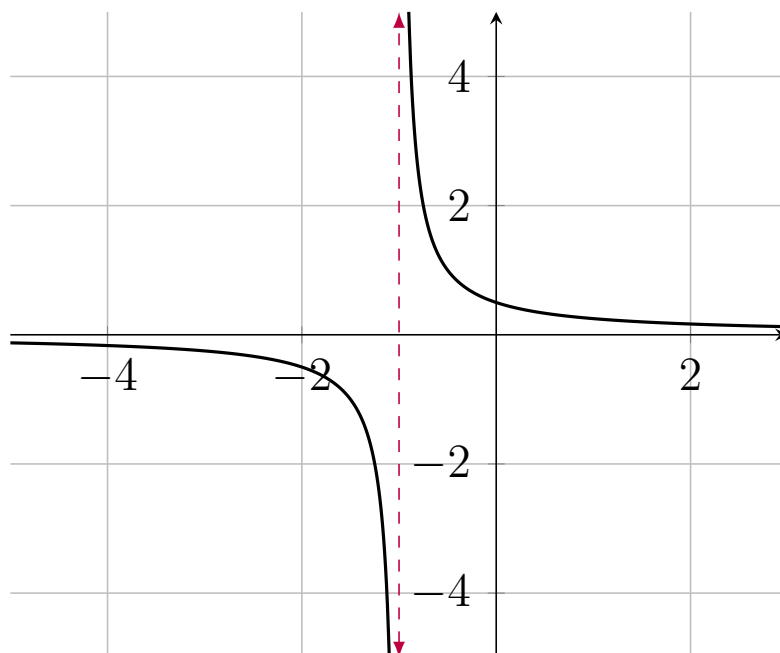
has a vertical asymptote at $x = c$.

Example 5. Determine all vertical asymptotes of the graph of

$$(a) f(x) = \frac{1}{2(x+1)} \quad (b) f(x) = \frac{x^2+1}{x^2-1} \quad (c) f(x) = \cot x \frac{2}{(x-3)^2}$$

Solution.

- (a) When $x = -1$ the denominator of $f(x) = \frac{1}{2(x+1)}$ is 0 and the numerator is not 0. So, by the Theorem, we can conclude that the line $x = -1$ is a vertical asymptote as shown in the figure below.



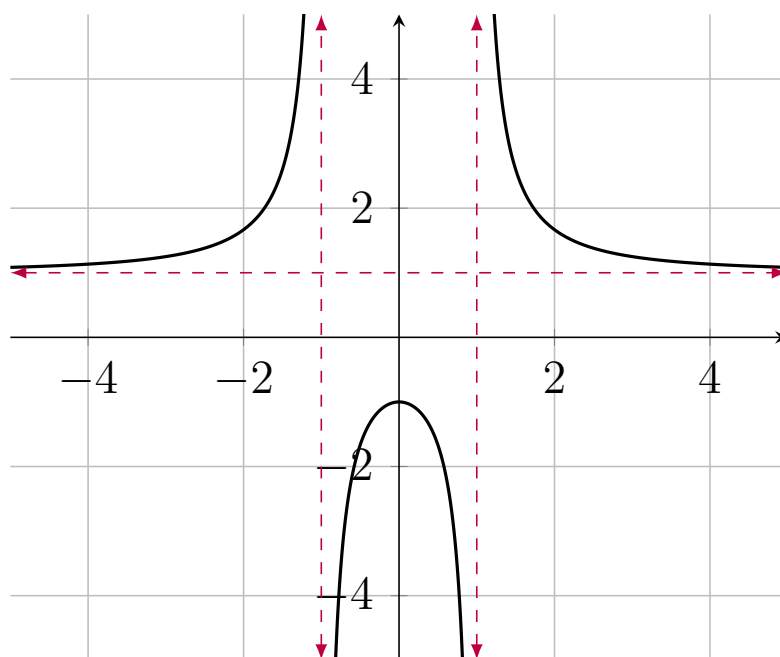
(b) By factoring the denominator as

$$f(x) = \frac{x^2 + 1}{x^2 - 1} = f(x) = \frac{x^2 + 1}{(x - 1)(x + 1)}$$

we see that the denominator is 0 at $x = 1$ and $x = -1$. Also, because the numerator is not 0 at these two points, we can apply the Theorem to conclude that the graph of f has two vertical asymptotes

$$x = -1 \text{ and } x = 1.$$

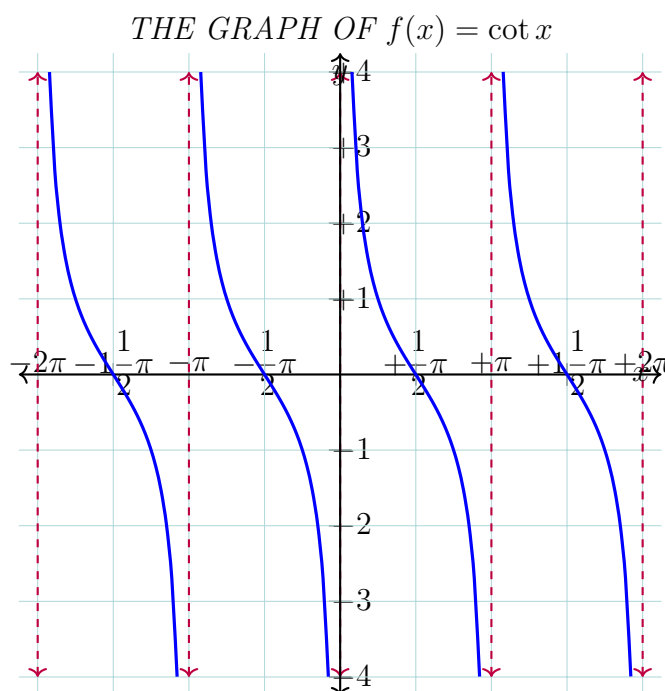
See the graph of f given below.



(c) By writing the cotangent function in the form

$$f(x) = \cot x = \frac{\cos x}{\sin x}$$

we can apply the theorem to conclude that vertical asymptotes occur at all values of x such that $\sin x = 0$. So, the graph of this function has infinitely many vertical asymptotes. These asymptotes occur at $x = n\pi$ where n is an integer. See the graph of $f(x) = \cot x$ given below to support our conclusion.



LIMITS AT INFINITY

We consider the "end behavior" of a function on an infinite interval.

Definition 6. *Intuitive Definition for Limit at Infinity*

1. Let f be a function defined on some interval (a, ∞) and $L \in \mathbb{R}$. Then

$$\lim_{x \rightarrow \infty} f(x) = L,$$

read as "the limit of $f(x)$ as x approaches ∞ is L ," means that the values of $f(x)$ becomes arbitrarily close to L as x takes on large positive numbers.

2. Let f be a function defined on some interval $(-\infty, a)$ and $L \in \mathbb{R}$. Then

$$\lim_{x \rightarrow -\infty} f(x) = L,$$

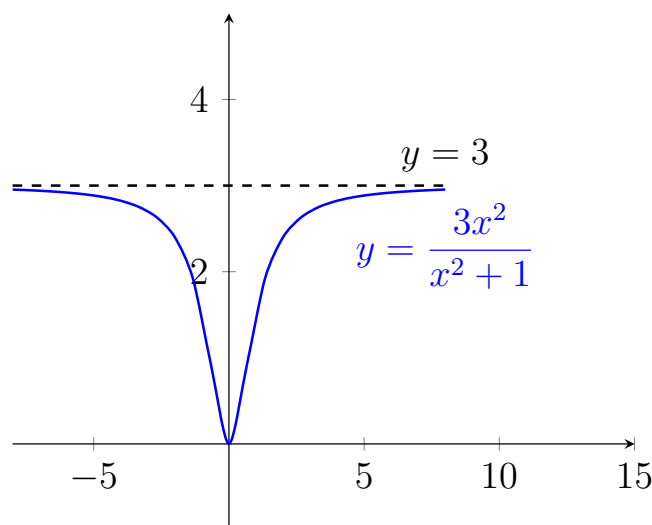
read as "the limit of $f(x)$ as x approaches $-\infty$ is L ," means that the values of $f(x)$ becomes arbitrarily close to L as x takes on large negative numbers.

Remark 7. The basic rules of limit, the substitution theorem and squeezing theorem are valid for limits at infinity when $x \rightarrow c$ is replaced by $x \rightarrow \infty$ or $x \rightarrow -\infty$.

Example 8. Find limits at infinity of the function $f(x) = \frac{3x^2}{x^2 + 1}$.

Solution.

Consider the graph of the function given below.



Graphically, we can see that the values of $f(x)$ appear to approach 3 as x increases without bound or decreases without bound.

Numerically, we can construct the table as shown below.

x	$-\infty \leftarrow$	-1000	-100	-10	-1	0	1	10	100	1000	$\rightarrow \infty$
$f(x)$	$3 \leftarrow$	2.999997	2.9997	2.9703	1.5	0	1.5	2.9703	2.9997	2.999997	$\rightarrow 3$

The table suggests that the value of $f(x)$ approaches 3 as x increases without bound ($x \rightarrow \infty$). Similarly, $f(x)$ approaches 3 as x decreases without bound ($x \rightarrow -\infty$).

Therefore, limits at infinity are given as

$$\lim_{x \rightarrow -\infty} f(x) = 3 \quad \text{Limit at negative infinity}$$

and

$$\lim_{x \rightarrow \infty} f(x) = 3 \quad \text{Limit at positive infinity}$$

Theorem 9 (Limits at Infinity). If r is a positive rational number and c is any real number, then

$$\lim_{x \rightarrow \infty} \frac{c}{x^r} = 0.$$

Furthermore, if x^r is defined when $x < 0$, then

$$\lim_{x \rightarrow -\infty} \frac{c}{x^r} = 0.$$

Example 10 (Finding a Limit at Infinity). Find the limits: $\lim_{x \rightarrow \infty} \left(2 + \frac{3}{x^2}\right)$ and $\lim_{x \rightarrow -\infty} \left(2 + \frac{3}{x^2}\right)$.

Solution. Let $f(x) = 2 + \frac{3}{x^2}$.

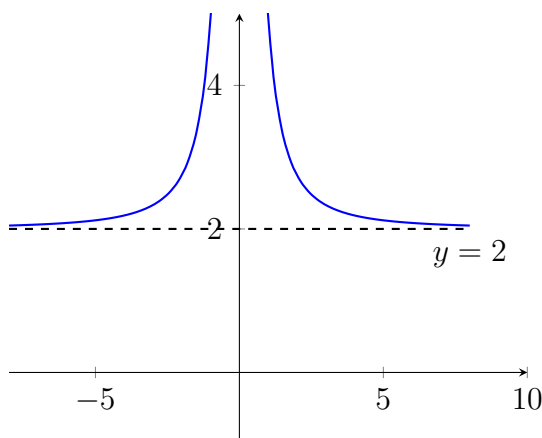
I. Numerical approach

x	$-\infty \leftarrow$	-1000	-100	-10	-1	0	1	10	100	1000	$\rightarrow \infty$
$f(x)$	$2 \leftarrow$	2.000003	2.0003	2.03	5	$\nexists$	5	2.03	2.0003	2.000003	$\rightarrow 2$

From the table we conclude that

$$\lim_{x \rightarrow -\infty} \left(2 + \frac{3}{x^2}\right) = 2 \quad \text{and} \quad \lim_{x \rightarrow \infty} \left(2 + \frac{3}{x^2}\right) = 2$$

II. Graphical approach: Consider the graph of $f(x) = 2 + \frac{3}{x^2}$.



The graph suggests that the value of $f(x)$ approaches 2 as x increases without bound ($x \rightarrow \infty$). Similarly, $f(x)$ approaches 2 as x decreases without bound ($x \rightarrow -\infty$).

Horizontal Asymptote

Definition 11 (Horizontal Asymptote). The horizontal line $y = L$ is called a **horizontal asymptote** of the graph of f , when

$$\lim_{x \rightarrow -\infty} f(x) = L \quad \text{or} \quad \lim_{x \rightarrow \infty} f(x) = L.$$

Example 12. Evaluate the following limits at infinity.

1. $\lim_{x \rightarrow \infty} \left(\frac{1}{x^3} - \frac{2}{x^2} + 3 \right)$

2. $\lim_{x \rightarrow \infty} \left(\frac{2x - 3}{x + 2} \right).$

Solution.

In class

Example 13. Find horizontal asymptotes of the graph of the following functions.

(a) $f(x) = \frac{2x + 5}{3x + 7}$

(b) $f(x) = \frac{x + 5}{2x^2 - 1}$

Solution.

In class

Example 14. Find $\lim_{x \rightarrow \infty} \left(5 - \frac{2}{x^2}\right)$.

Solution.

In class

Example 15 (A Comparison of Three Rational Functions). *Find each limit.*

$$(a) \lim_{x \rightarrow -\infty} \frac{2x + 5}{3x^2 + 1} \quad (b) \lim_{x \rightarrow -\infty} \frac{2x + 5}{3x^2 + 1} \quad (c) \lim_{x \rightarrow -\infty} \frac{2x^3 + 5}{3x^2 + 1}$$

Solution.

In class.

Guidelines for finding limits at $\pm\infty$ of rational functions

1. If the degree of the numerator is less than the degree of the denominator, then the limit of the rational function is 0.
2. If the degree of the numerator is equal to the degree of the denominator, then the limit of the rational function is the ratio of the leading coefficients.
3. If the degree of the numerator is greater than the degree of the denominator, then the limit of the rational function does not exist.

Infinite Limits at Infinity

Many functions do not approach a finite limit as x increases (or decreases) without bound.

Definition 16. *Infinite Limits at Infinity*

1. Let f be a function defined on the interval (a, ∞) .
 - (a) The statement $\lim_{x \rightarrow \infty} f(x) = +\infty$ means that $f(x)$ increases without bound ($f(x) \rightarrow \infty$), when x increases without bound ($x \rightarrow \infty$).
 - (b) The statement $\lim_{x \rightarrow \infty} f(x) = -\infty$ means that $f(x)$ decreases without bound ($f(x) \rightarrow -\infty$), when x increases without bound ($x \rightarrow \infty$).
2. Let f be a function defined on the interval $(-\infty, a)$.
 - (a) The statement $\lim_{x \rightarrow -\infty} f(x) = +\infty$ means that $f(x)$ increases without bound ($f(x) \rightarrow \infty$), when x decreases without bound ($x \rightarrow -\infty$).
 - (b) The statement $\lim_{x \rightarrow -\infty} f(x) = -\infty$ means that $f(x)$ decreases without bound ($f(x) \rightarrow -\infty$), when x decreases without bound ($x \rightarrow -\infty$).

Example 17. Using numerical approach find the following limit.

$$(a) \lim_{x \rightarrow \infty} x^3 \quad (b) \lim_{x \rightarrow -\infty} x^3$$

Solution. Numerically, we can construct the table as shown below.

x	$-\infty \leftarrow$	-1000	-100	-10	-1	0	1	10	100	1000	$\rightarrow \infty$
$f(x) = x^3$	$-\infty \leftarrow$	-10^9	-10^6	-1000	-1	0	1	1000	10^6	10^9	$\rightarrow \infty$

The table suggests that the value of

- i. $f(x)$ increases without bound ($f(x) \rightarrow \infty$) as x increases without bound ($x \rightarrow \infty$).
- ii. $f(x)$ decreases without bound ($f(x) \rightarrow -\infty$) as x decreases without bound ($x \rightarrow -\infty$).

Therefore, infinite limits at infinity are given as

$$\lim_{x \rightarrow -\infty} x^3 = +\infty \quad \text{and} \quad \lim_{x \rightarrow -\infty} f(x) = -\infty$$

Example 18. Find each limit.

$$(a) \quad \lim_{x \rightarrow \infty} \frac{2x^2 - 4x}{x + 1} \qquad (b) \quad \lim_{x \rightarrow -\infty} \frac{2x^2 - 4x}{x + 1}$$

Solution.

In class

A Summary of Curve Sketching

We have studied the following concepts that are useful in analyzing the graph of a function.

1. Domain and range
2. Intercepts
 - (a) x -intercepts
 - (b) y -intercepts
3. Symmetry
 - (a) Symmetric with respect to the origin
 - (b) Symmetric with respect to y -axis
4. Continuity
5. Vertical asymptotes
6. Differentiability
7. Monotonicity (First Derivative Test)
 - (a) Identifying intervals on which f is increasing
 - (b) Identifying intervals on which f is decreasing
8. Relative extrema
 - (a) Relative maximum
 - (b) Relative minimum
9. Concavity
 - (a) Concave Upward

(b) Concave downward

10. Points of inflection

11. Horizontal asymptotes

12. Infinite limits at infinity

Example 19 (Sketching the Graph of a Rational Function). *Analyze and sketch the graph of*

$$f(x) = \frac{2(x^2 - 9)}{x^2 - 4}.$$

Solution

In class

Example 20 (Sketching the Graph of a Polynomial Function). *Analyze and sketch the graph of*

$$f(x) = x^4 - 12x^3 + 48x^2 - 64x.$$

Solution

In class

Example 21 (Sketching the Graph of a Radical Function). *Analyze and sketch the graph of*

$$f(x) = 2x^{\frac{5}{3}} - 5x^{\frac{4}{3}}$$

Solution

In class